

***THREEFOLD SOURCES SOUGHT FOR
PRODUCT(S) MADE IN AMERICA & SMALL BUSINESS MANUFACTURER(S)
THIS NOTICE IS FOR MARKET RESEARCH PURPOSES ONLY
NOT A REQUEST FOR PROPOSAL OR SOLICITATION**

Sources Sought Number: 75F40126SS132448

Information regarding any technical point of contact will not be provided and no appointments for presentations will be made.

Notice of Intent

Responses to this Sources Sought announcement will assist the Government in determining whether any future requirements similar to this one should be set aside for small business, made available to full and open competition or procure through sole-source acquisition procedures. Interested Parties shall respond with capability statements which are due by email to howard.nesmith@fda.hhs.gov by **1:00 PM** (Central Time), **March 21, 2026, Referencing 75F40126SS132448.**

The U.S. Food and Drug Administration (FDA), Center for Devices and Radiological Health (CDRH), requires the acquisition of an Electro-Mechanical Dynamic Load Frame System and post-warranty Service and Maintenance Agreement as specified herein.

***Part One**

Product(s) Made In America

Executive Order 14005 of January 25, 2021, was issued Ensuring the Future is Made in All of America by All of America's Workers, which directs Federal agencies to maximize the use of goods, products, and materials produced in, and services offered in, the United States through their financial assistance awards and procurements.

In compliance with Executive Order 14005 the FDA is seeking potential provider(s) who can provide and install an Electro-Mechanical Dynamic Load Frame System which is **Made in America**. Regardless of North American Industry Classification System (NAICS) Code Business Size Standards all interested parties who can provide the product(s) described herein **Made in America** may respond to this Notice.

***Part Two**

Small Business Manufacturer(s)

Potential provider(s) considered a **Small Business** under North American Industry Classification System (NAICS) 334516 – Analytical Laboratory Instrument Manufacturing; with a Small Business Size Standard of 1,000 employees and is capable of providing the FDA with the product(s) described herein and **Made in America** may respond to this Notice.

***Part Three**

Product(s) Made Outside America

Potential provider(s) no matter their North American Industry Classification System (NAICS) size who can provide the product(s) described herein which is made **Outside America** may also respond to this Notice.

All responses shall be submitted via email to Howard.Nesmith@fda.hhs.gov and provide:

- Proposed Product Information and Description
- Capability Statement which indicates proposed product(s) is Made in America
- Business Name
- Unique Entity ID (SAM) Number
- Business Contact Information to include point of contact name, mailing address, phone number and email address
- NAICS Small Business Program Representations (Small or Large)
- **Country of Origin of proposed product(s) if Made Outside America**

Background:

The FDA/CDRH, currently has an electro-dynamic load frame system with a 3,000 N capacity. This capacity is appropriate for small medical devices and/or small material samples. However, larger medical devices and/or material samples require increased load capacity. As such, CDRH is limited in the size and materials we can currently test on the existing electro-dynamic load frame. Investing in a new electro-dynamic load frame system with a 10,000 N capacity would greatly expand CDRH's ability to test medical devices and materials.

General Requirements/Specifications:

Any catalog number, manufacturer brand name, or proprietary description used in the Minimum Technical Specifications are used as descriptive only, not restrictive, and used to indicate type, function, or quality desired.

Electro-Mechanical Dynamic Load Frame System

- Shall be floor-standing and a dual support column design.
- Shall be of an oil-free (i.e., not hydraulic) design to minimize contamination potential of specimens and to minimize maintenance.
- Shall have a method to ensure the frame is level with the floor (e.g., 4 leveling feet or foot plates). It shall also have an option to secure the load frame to the floor.
- Due to the agency's constraints such as lab dimensional constraints and moving logistics, the up righted load frame dimensions shall be: Height (defined as perpendicular to ground) shall be capable of being under 92" (2337 mm). This is the lab door height. The load frame height shall be no more than 140" (3556 mm).
 - Width shall be under 48" (1219 mm).
 - Depth shall be under 48" (1219 mm).
 - These dimensions do not include mounted touchscreen operator display(s), dashboard(s), tablet(s), or jog control pendant(s). With mounted operator tablet(s), dashboard(s), or jog control pendant(s) the system shall be no more than 140" x 65" x 60" (3556 mm x 1651 mm x 1524 mm) in height, width, and depth.
- Shall be no more than 3,200 lbs. (1451 kg) without accessories.
- Shall have a space between columns (i.e., daylight opening) of 16" (406 mm) or more. This space is needed to facilitate a multitude of samples, sensors, and other experimental equipment.
- Shall have an adjustable vertical test space capable of 30" (762 mm) or more.
- crosshead shall have the ability to be lowered so the vertical test space is at 4" (102 mm) or less from the baseplate.

- Shall have an electrically powered crosshead lift system with manual lever clamps. Interface with this motion system shall be controllable with a control panel or jog pendent. This electromechanical positioning system should have the ability to lock (e.g., clamp and tighten) the crosshead in place to the columns.
- Main actuator shall be housed in the crosshead on the upper side of the test space.
- Load cell for the load frame shall not be incorporated into the moving actuator.
- Load frame system shall have mounting slots and/or threaded holes on the base plate so the user can attach accessories to the load frame base plate.
- Power requirements (without computer and accessories) shall be compatible with 208-240 V, single phase, 50-60 Hz, 32 Amp outlets.
- Force or torsion capacity shall dictate the types and sizes of materials and/or components it can test. Therefore, shall have:
 - A static force capacity of $\pm 1,500$ lbf (6.7 kN) or more.
 - A dynamic force capacity of $\pm 2,200$ lbf (9.8 kN) or more.
 - A dynamic torque capacity of ± 800 lbf-in (100 N-m) or more.
 - A load and torque measurement accuracy: $\pm 0.75\%$ of indicated load or torque, or $\pm 0.0075\%$ of load cell capacity, whichever is greater.
- Motion capabilities shall dictate the type of displacements and/or strains it can impose on a material or assembly. Therefore, the load frame shall have:
 - Total (peak to peak) linear stroke capacity of at least 2" (50.8 mm) or more.
 - Total rotation capacity of at least $\pm 120^\circ$ or more.
 - Since the load frame will have the ability to perform dynamic (e.g., fatigue) experiments, the speed of the system shall be adjustable up to at least 100 Hz.
- Shall have an operating temperature range of +12 to +30 °C or wider.
- Shall have an allowable operating humidity of 10% to 90% (non-condensing).
- Shall have features (i.e., safety interlocks) that protects the load frame and specimens when the operator moves the crosshead. One of these features shall be to detect and prevent high forces on the sample during operator setup. Shall also have:
 - Safety interlocks and emergency stop capabilities.
 - Optional safety guard compatibility.
- Shall come with a controller (e.g., jog pendent) that can be either handheld or attached to the load frame so the user can easily control the system during test specimen setup.
 - The controller shall be able to jog the crosshead for general and precision positioning.
 - The controller shall have an emergency stop button to safely stop the system when triggered.
- Shall have two controllable axes of motion
 - Axial and torsional
 - Capability to do bi-axial control.
 - Capability to do bi-modal and tri-modal control.
- Data acquisition shall have:
 - At least 8 kHz continuous synchronous data acquisition on all channels.
 - At least 3 closed loop control and data acquisition channels for position, digital
 - At least 3 closed loop control and data acquisition channels for angular position, digital angular position, torque and on the torsional axis.
 - At least 2 closed-loop control and data acquisition channels for other input sensors.

- At least 2 analogue data inputs and 4 analogue data outputs.
 - At least 8 digital inputs and 8 digital outputs.
- Shall come with one compatible (1) load cell:
 - Shall have a usable range from maximum capacity down to 1/200th of the maximum capacity or better.
 - Accuracy shall be $\pm 0.75\%$ or better of the output reading from the maximum capacity or $\pm 0.0075\%$ of its maximum capacity, whichever is better.
 - The overload capacity shall be at least 125% without being damaged and 250% without mechanically failing the load cell. Expensive repairs and downtime are costly, and this reduces that potential cost and downtime.
 - System shall be temperature compensated.
 - System setup shall have automated features to automatically recognize when connected (i.e., manual selection of calibration/data files is not necessary).
 - Shall be suitable for different testing modes such as cyclic and completely reversed loading, quasi-static tension, and quasi-static compression.
 - Dynamic fatigue testing systems shall have significant mass in the moving testing assembly and shall have automated inertial compensation to account for this (i.e., remove amplitude and phase errors).
 - Shall be able to survive at least 1 billion completely reversed cycles at max load.
 - Shall comply with ASTM E4, ISO 7500-1 class 0.5.
- Shall come with system controller that is capable of:
 - Mouse and keyboard input.
 - Transfer data via USB flash drives.
- System shall not require internet connectivity to operate as described in these salient requirements.
- System shall also:
 - Be compatible with dynamic testing software supporting waveform generation.
 - Have real-time calculation capabilities.
 - Have advanced control modules for frequency sweeps and amplitude control
 - Have operator security features with password and/or PIN access control.
 - Have test review and data analysis capabilities.
- System shall have common, non-proprietary, data transfer ports (e.g., USB ports) to transfer sample data (e.g., results of tensile testing, compression testing) acquired from the load frame system.
- System shall be able to export data in common, non-proprietary, data formats (e.g., *.txt, *.csv, *.xls, *.xlsx).
- System shall have the following **optional** accessories:
 - A safety shield / guard that protects the user from potentially dangerous sample fragments that are released during testing.
 - An extension kit which allows the actuator reach to be lengthened so that water baths or thermal ovens may be used. This kit should allow the length of the pull rod actuator to be lengthened at least 6 to 10”.
 - One (1) set of compression platens (at least 3” in diameter) with LVDT extensometer mounting capabilities. The compression platens shall be able to sustain the load frame’s linear and torsional loads.
 - One (1) LVDT deflection sensor that is compatible with the compression platens. The deflection sensor shall have an operation range of ± 0.5 mm or greater.

- One (1) set fatigue rated manual mechanical grips which are rated for the system's full linear and torsional load capacity. The grips shall come with one full set of flat serrated wedges which accommodate a specimen engagement thickness of up to 0.25", an engagement width of at least 1" wide, and an engagement height of at least 1".
- One (1) set fatigue rated pneumatically operated mechanical grips which are rated for the system's full linear and torsional load capacity. The grips shall come with one full set of flat serrated wedges which accommodate a specimen engagement thickness of up to 0.25", an engagement width of at least 1" wide, and an engagement height of at least 1".
- One (1) non-contact video extensometer capable of being strain verified to ASTM E83.
- One compatible (1) dynamic extensometer that has a gauge length between 0.25 to 1.00" and be able to extend/retract at least $\pm 10\%$ strain. This strain gauge extensometer shall be capable of being verified to ASTM E83.
- One compatible (1) static extensometer that has a gauge length between 0.25 to 0.75" and be able to extend/retract at least $\pm 10\%$ strain. This strain gauge extensometer shall be capable of being verified to ASTM E83.
- System shall have complete manuals and instructions in digital format so that the user can operate the load frame system within the salient requirements stated in the award. These manuals and instructions shall be available on the vendors website as long as they are accessible and downloadable by the agency. All manuals and instructions for the system shall be in clear English.
- Shall include on-site installation to include full system hardware, peripherals, software, qualification and training.
- Shall provide onsite software integration, and training on software fundamentals and test method development.
- Shall include on-site force verification of the load cell(s) and load frame at the final installation site, after installation.
- Shall include on-site crosshead speed and displacement verification (e.g., ASTM E2658, ASTM E2309) under no load and non-cyclic conditions at the final installation site, after installation.
- Shall provide onsite instruction and consultancy to at least (2) FDA employees. The training shall take no more than eight (8) days to complete.
- Shall include at minimum of one (1) year comprehensive Warranty, Maintenance and Technical Support from the date of installation and acceptance.
 - Shall cover all equipment repair costs within the warranty time period that are the result of general in-specification use.

Minimum Post Warranty Service/Maintenance Requirements:

- All maintenance and repair activities shall be performed by formally trained and certified Original Equipment Manufacturer (OEM) technicians/engineers.
- All repairs and PM services shall be performed following OEM specifications, manuals, and service bulletins, using OEM replacement parts, components, subassemblies, etc.

- All Corrective/Remedial Service(s) and Preventive Maintenance (PM) shall be inclusive of shipping, labor, travel, replacement parts, subassemblies, components, or modules necessary for repairs, etc.
- PM Service shall include at least one (1) on-site visit for routine instrument calibration/certification per year inclusive of all consumable parts required to complete the PM.
- Any part(s) that cannot be repaired on-site, shipping arrangements and material (e.g., boxes/crates) shall be provided without extra cost.
- Corrective/Remedial Repair visits shall be provided within 2-3 business days of call for service (for issues that can't be resolved remotely via phone or Internet).
- Service plan shall include direct access to OEM call center accessible by FDA Contracting Officer's Representative (COR), Technical Representative (TR) and/or System Operator Personnel for technical assistance which is staffed by technicians/engineers capable of providing high level troubleshooting expertise.
- Service plan shall include OEM technical developments, repair procedure bulletins, and unlimited software, firmware, and application updates.
- Technical support/assistance. (e.g., telephone-based, email-based, website-based, etc.) shall be provided Monday - Friday (excluding Federal Holidays) during normal business hours and shall begin on a solution to the problem within 8 business hours of contact for assistance.
- Deliverables: Service Records and Reports
The Contractor shall, commensurate with the completion of each service call or preventative maintenance visit, provide the end-user of the equipment and the Contracting Officer with a copy of a field service report/ticket identifying the equipment name, manufacturer, model number, and serial number of the equipment being serviced/repared and detailing the reason for the service call, a detailed description of the work performed, the test instruments or other equipment used to affect the repair or otherwise perform the service, the name(s) and contact information of the technician who performed the repair/service, and for information purposes, the on-site hours expended and parts/components replaced.

FOB Destination & Place of Performance:

U.S. Food and Drug Administration
Center for Devices and Radiological Health
10903 New Hampshire Ave
Silver Spring, MD 20993

Respondents shall furnish sufficient technical information necessary for the Government to conclusively determine that the offered product/services meet the General Requirements/Specifications identified herein. Though the target audience is Small Business Manufacturers or Small Businesses capable of supplying services of a Small Business, all interested parties may respond. At a minimum, responses shall include the following:

- Business name, DUNS number, business address, business website, business size status (i.e., SB, VOSB, SDVOSB, HUBZone SB, SDB, WOSB, LB), point of contact name, mailing address (if different from business address), phone number and email address (Provide this same information if responding to provide a product manufactured by another firm);

- Sufficient descriptive literature that unequivocally demonstrates that offered service can meet the herein General Requirements/Specifications. All descriptive material necessary for the government to determine whether the offered meets General Requirements/Specifications including technical specifications, descriptive material, literature, brochures and other information, which demonstrates the capabilities of the respondent to meet the General Requirements/Specifications;
- If applicable, identification of the respondent's GSA Schedule contract(s) by Schedule number and SINs that are applicable to this potential requirement. Detailed list of items included and not included on the GSA contract to meet this General Requirements/Specifications;
- If a large business, identify the subcontracting opportunities that would exist for small business concerns;
- Though this is not a request for quote, informational pricing is encouraged.

The government is not responsible for locating or securing any information, not identified in a response.

The Government encourages any comments and/or suggestions from any interested party, regarding the specifications. While the Government will not respond directly to your comments and/or suggestions; we will consider them as we finalize the specifications in preparation for the forthcoming solicitation.

Disclaimer and Important Notes

This notice does not obligate the Government to award a contract or otherwise pay for the information provided in response. The Government reserves the right to use information provided by respondents for any purpose deemed necessary and legally appropriate. Any organization responding to this notice shall ensure that its response is complete and sufficiently detailed to allow the Government to determine the organization's qualifications to perform the work. Respondents are advised that the Government is under no obligation to acknowledge receipt of the information received or provide feedback to respondents with respect to any information submitted. After a review of the responses received, a pre-solicitation synopsis and/or a solicitation may be published in SAM.gov. However, responses to this notice will not be considered adequate responses to a solicitation.

Confidentiality

No proprietary, classified, confidential, or sensitive information shall be included in your response. The Government reserves the right to use any non-proprietary technical information in any resultant solicitation(s).